```
Last login: Tue Jul 14 16:25:18 on ttys020
(base) sneharajalakshmipalanisamy@Rajalakshmis-MacBook-Air ~ % cd Downloads/eeai_test
(base) sneharajalakshmipalanisamy@Rajalakshmis-MacBook-Air eeai_test % source venv/bin/activate
(venv) (base) sneharajalakshmipalanisamy@Rajalakshmis-MacBook-Air eeai_test % python3 code.py --input-path input_test.npy --
output-path my_output.jsonl
/Users/sneharajalakshmipalanisamy/Downloads/eeai_test/venv/lib/python3.11/site-packages/tensorflow/lite/python/interpreter.p
y:457: UserWarning:     Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
    TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
    See the [migration guide](https://ai.google.dev/edge/litert/migration)
    for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
  frame 0/5000  (0s)
  frame 1000/5000  (1s)
  frame 2000/5000  (2s)
  frame 3000/5000  (4s)
  frame 4000/5000  (5s)
Inference pass done in 6s
Wrote 5000 frames -> my_output.jsonl
(venv) (base) sneharajalakshmipalanisamy@Rajalakshmis-MacBook-Air eeai_test % python3 evaluate_performance.py --gt-path outp
ut_test.jsonl --pred-path my_output.jsonl
Output validation PASSED.
Loading files...
  Ground truth : 5000 frames   (output_test.jsonl)
  Predictions  : 5000 frames   (my_output.jsonl)
  Common frames                  : 5000

============================================================
  DETECTION SCORE — F1
============================================================
  Matching threshold  : 1.0 m
  Frames evaluated    : 5000
  TP / FP / FN        : 9955 / 16 / 45
  Precision           : 0.9984
  Recall              : 0.9955
  F1 Score            : 0.9969

------------------------------------------------------------
LOCALISATION ERROR  (matched pairs only)
```

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──────────────────────────────────────────────────────────────
  Matched pairs       : 9955
  RMSE                : 0.2457 m
  MAE                 : 0.2062 m
  Median error        : 0.1816 m
  P90 error           : 0.3627 m

──────────────────────────────────────────────────────────────
PERSON COUNT
──────────────────────────────────────────────────────────────
  Count MAE           : 0.0058 persons / frame
  Count accuracy      : 99.42%  (4971 / 5000 frames)
============================================================
JSON_RESULT: {"f1": 0.9969, "precision": 0.9984, "recall": 0.9955, "rmse": 0.2457}
(venv) (base) sneharajalakshmipalanisamy@Rajalakshmis-MacBook-Air eeai_test % python3 evaluate_constraint.py --model-path mo
del.tflite
============================================================
  ESP32-S3 CONSTRAINT CHECK
============================================================
  Model : model.tflite
  Size  : 577.9 KB

CHECK 1  TFLite model validity
/Users/sneharajalakshmipalanisamy/Downloads/eeai_test/venv/lib/python3.11/site-packages/tensorflow/lite/python/interpreter.p
y:457: UserWarning:     Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
    TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
    See the [migration guide](https://ai.google.dev/edge/litert/migration)
    for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
  [PASS] Model loads and allocates — input [1, 6, 3, 60, 6] (int8), output [1, 4, 3] (int8)

CHECK 2  Model file size (limit: 800 KB Flash)
  [PASS] 577.9 KB < 800 KB

CHECK 3  Activation arena estimate (limit: 300 KB SRAM)
         Activation tensors: 212.4 KB  (TFLM overlaps non-simultaneous buffers — actual arena ≤ this)
  [PASS] ~212.4 KB < 300 KB — fits in SRAM
```

```
CHECK 4  Operations (LSTM / GRU / RNN / CUSTOM / Flex not supported)
  [PASS] No forbidden operations detected

CHECK 5  Quantization (INT8 strongly recommended for ESP32-S3)
  [PASS] Full INT8 quantization — optimal for ESP32-S3

============================================================
  RESULT: PASS
============================================================
  All hard constraints pass.
============================================================

JSON_RESULT: {"passed": true, "file_size_kb": 577.9, "arena_kb": 212.4}
(venv) (base) sneharajalakshmipalanisamy@Rajalakshmis-MacBook-Air eeai_test %
```